东 南 大 学 考 试 卷（ B 卷）

课程名称 离散数学 考试学期 2020-2021-3 得分

适用专业 计算机类 考试形式 闭卷 考试时间长度 120 分钟

（开卷、半开卷请在此写明考试可带哪些资料）

题目	1	2	3	4	5	6	7	8	9	10	总分
得分											
批阅人											

自觉遵守考场纪律 如考试作弊 此答卷无效

姓名

学号

线 封 密

1. Choice questions （ 2 points for each question ）

(1) Which of the following compound propositions is not a tautology? ( )

A. $(p \wedge q) \rightarrow q$

B. $p \wedge (p \vee q) \leftrightarrow p$

C. $(\neg p \wedge (p \rightarrow q)) \rightarrow \neg q$

D. $\neg (p \rightarrow q) \rightarrow \neg q$

(2). $\overline{A \cup (B \cap C)}$ = ( ).

A. $(\overline{C} \cup \overline{B}) \cap \overline{A}$.

B. $\overline{A} \cup \overline{B \cap C}$.

C. $\overline{A} \cap \overline{B \cup C}$.

D. $\overline{A \cup B} \cap \overline{C}$.

(3). Which of these relations on the set of all people is equivalence relation ( ).

A. {(a, b) | a and b have the same parents}

B. {(a, b) | a and b share a common parent}

C. {(a, b) | a and b have met}

D. {(a, b) | a and b speak a common language}

(4). For a group <G, *>, the operation * must satisfy ( ).

A. commutative law B. exists zero element

C. idempotent law D. every element has inverse element

(5). Which of the following operations does NOT satisfy associative law? ( )

A. a*b=a+b+ab B. a*b=a mod b

C. a*b=$\sqrt[3]{a^3+b^3}$ D. a*b=min(a, b)

(6). For G = _____, G is an Euler graph. ( )

A. $K_{5,6}$ B. $K_{5,5}$

C. K_5 D. K_6

(7). For G = _____, G is a Hamiltonian graph. ( )

A. $K_{2,8}$ B. $K_{3,7}$

C. $K_{4,6}$ D. $K_{5,5}$

2. Completion （2 points for each question）

(1). Let *C(x)*, *D(x)*, and *J(x)* be the statements "x is a student in the class," "x has studied discrete mathematics," "x has studied Java," respectively. Suppose that the domain consists of all people. Express each of these statements using quantifiers; logical connectives; and *C(x)*, *D(x)*, and *J(x)*.

a) No student in the class has studied Java.

_____.

b) Some students in the class have studied discrete mathematics and Java.

_____.

(2). The equivalence class of 0 for congruence modulo 4 in set Z is _____ _____, and the equivalence class of 3 for congruence modulo 4 in set Z is _____ _____,

(3). Suppose there is a cyclic group G of order 21, which can be generated by a generator g, i.e. G={1, g, g^2, …, g^{20}}. Then, the sub-group H generated by <g^7> is _____. (the identity element can be denoted by 1)

(4). For a power set P(A) of a set A and the set operations ∩, ∪, and ⊕, the identity elements of the above three operations are _____, respectively.

(5). In K_7, there are _____ edges and the degree of each node is _____.

3. Find the Disjunctive Normal Form of the following formula. (6 points)

$((p \vee q) \rightarrow r) \rightarrow p$

4. Translate the following sentence into predicate logic and use the rules of inference to construct a valid argument. (10 points)

"Linda, a student in this class, owns a red convertible. Everyone who owns a red convertible has gotten one speeding ticket. Therefore, someone in this class has gotten a speeding ticket."

5. What is the definition of transitive closure? Find the zero-one matrix of the transitive closure of the relation R where $M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$. (9 points)

6. Let R be the equivalence relation in X, and S be defined as

$$S = \{< a, b > | \exists c \in X, \text{make} < a, c > \in R \wedge < c, b > \in R\}.$$

Show that S is also an equivalence relation in X. （10 points）

7. Given set A={2, 3, 4, 5, 6, 8, 9, 12} and the division relation R, please answer the following questions. (10 points)

(1) Draw the Hasse diagram of poset <A, R>.

(2) Find the maximum and minimum element(s) of this poset.

(3) Is the maximum and minimum element(s) the greatest and least element?

(4) What is the upper bound of {2, 3} and what is its least upper bound?

(5) What is the lower bound of {8, 12} and what is its greatest lower bound?

8. Please explain the Lagrange theorem, and explain why a group of prime order must be a cyclic group using the Lagrange theorem. (9 points)

9. Suppose T is a tree and T has 2 vertices of degree 2, 3 vertices of degree 3, no vertex of degree more than 3. How many vertices of degree 1 does T have? （7 points）

10. Suppose that a connected bipartite planar simple graph with e edges and v vertices contains no simple circuits of length 4 or less. Show that $e \leqslant (3/2)v - 3$ if $v > 4$. (9 points)